Recitation 12 – IPv6, Network Layer & Link Layer

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Computer Networks

Table of Contents

1. IPv6
2. Network Layer Protocols
3. Network Layer vs Link Layer
4. Link Layer
5. Putting It All Together
6. Practice Problems
7. Collisions and CSMA/CD

Today

- **IPv6**
  - Motivation and Header Format
  - Changes from IPv4
  - Tunneling and Transition
- **Network Layer**
  - DHCP
  - ICMP
- **Link Layer**
  - Ethernet
  - ARP (Address Resolution Protocol)
- **Putting It All Together**
- **Practice Problems**
- **Collisions & CSMA/CD**

IPv6 – Motivation

Why IPv6?

Initial motivation: 32-bit IPv4 address space exhaustion

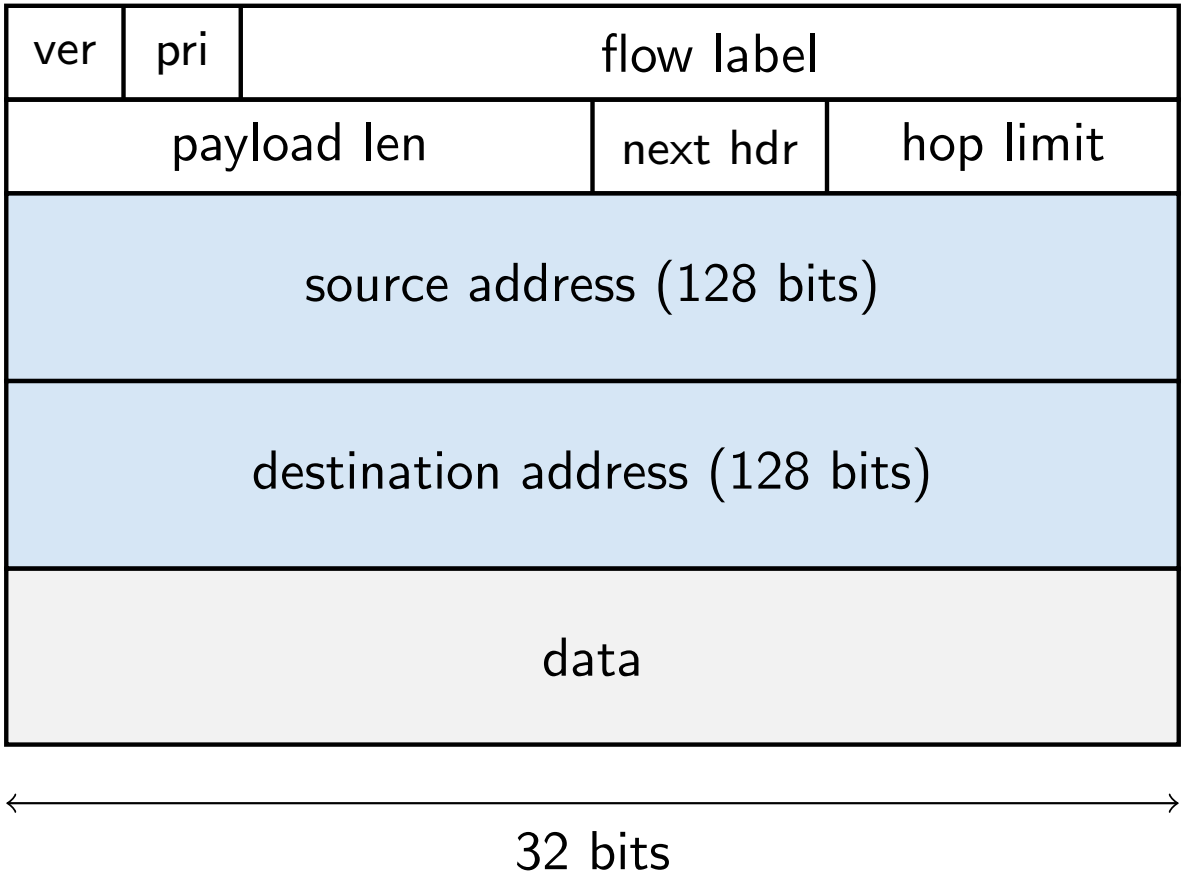
- Address space completely allocated (predicted 2008, occurred 2011-2019 by region)

Additional Motivations:

- **Simplified header format** – helps speed processing/forwarding
- **QoS support** – header changes to facilitate Quality of Service (prioritization of datagrams)
- **Anycast addressing** – route to "best" of several replicated servers (several servers with same IP)

IPv6 Datagram: Fixed-length **40 byte header** (vs. variable 20-60 bytes in IPv4)

IPv6 Header Format



- **Priority (pri):** identify priority among datagrams in flow
- **Flow Label:** identify datagrams in same "flow"
- **Next Header:** identify upper layer protocol (TCP=6, UDP=17)

Changes from IPv4

Changes from IPv4

- **Checksum** – removed entirely
  - Reduces processing time at each hop
  - Transport layer handles integrity
- **Fragmentation** – not allowed within network
  - Large datagrams not divided in transit
  - Source must handle MTU discovery

Additions in IPv6

- **ICMPv6** – new version of ICMP
  - "Packet Too Big" message
  - Multicast group management
  - Neighbor Discovery Protocol
- **128-bit addresses**
  - 2^{128} possible addresses
  - Written in hexadecimal

Transition from IPv4 to IPv6

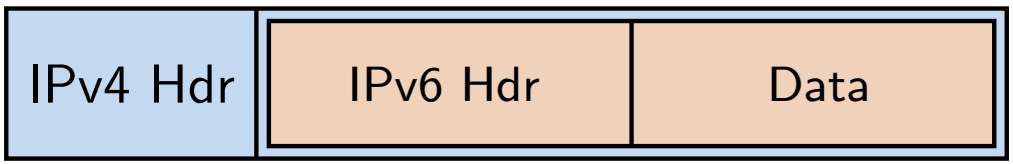
The Challenge

Not all routers can be upgraded simultaneously – no "flag days"!

- How will the network operate with **mixed IPv4 and IPv6** routers?

Solution: Tunneling

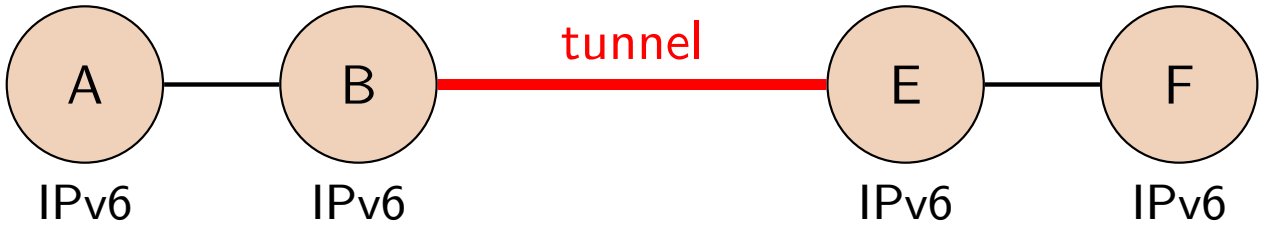
Tunneling: IPv6 datagram carried as **payload** in IPv4 datagram among IPv4 routers



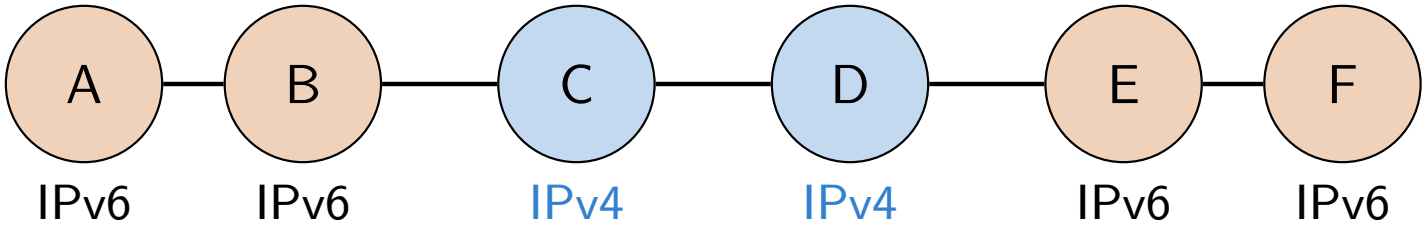
IPv6 packet encapsulated in IPv4

Tunneling – How It Works

Logical View:

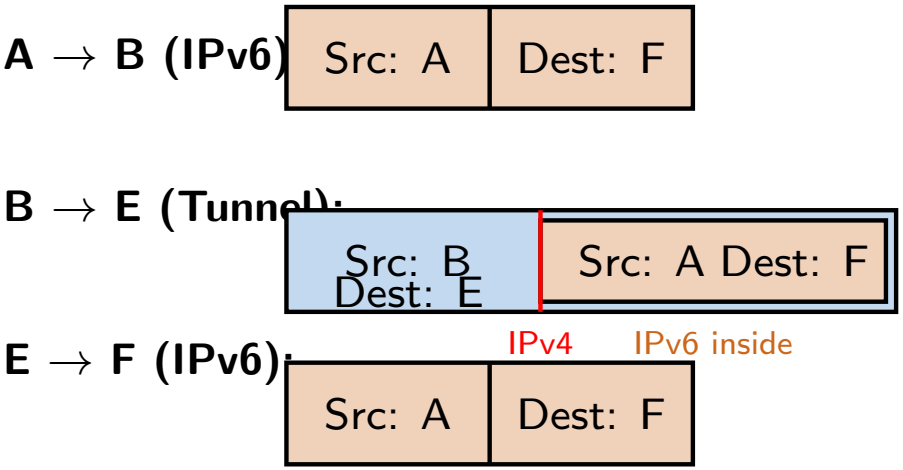


Physical View:



- **A-to-B:** Pure IPv6
- **B-to-E:** IPv6 encapsulated inside IPv4 (tunnel)
- **E-to-F:** Pure IPv6 (IPv4 header stripped)

Tunneling – Packet Transformation



Key Point: The original IPv6 source (A) and destination (F) addresses are **preserved** through the tunnel. Only the tunnel endpoints (B, E) appear in the IPv4 header.

IPv6 Adoption

Current Adoption Rates (2024)

IPv6 Capable Rate by Country:

- Israel: 62%
- United States: 55%
- India: 78%

Google Statistics: 42% of clients access Google services via IPv6 (Dec. 2024)

IPv6 Adoption

Long deployment time: 20+ years and counting!
Compare to application-level changes in same period: Facebook, smartphones, cloud computing...

DHCP – Dynamic Host Configuration Protocol

What is DHCP?

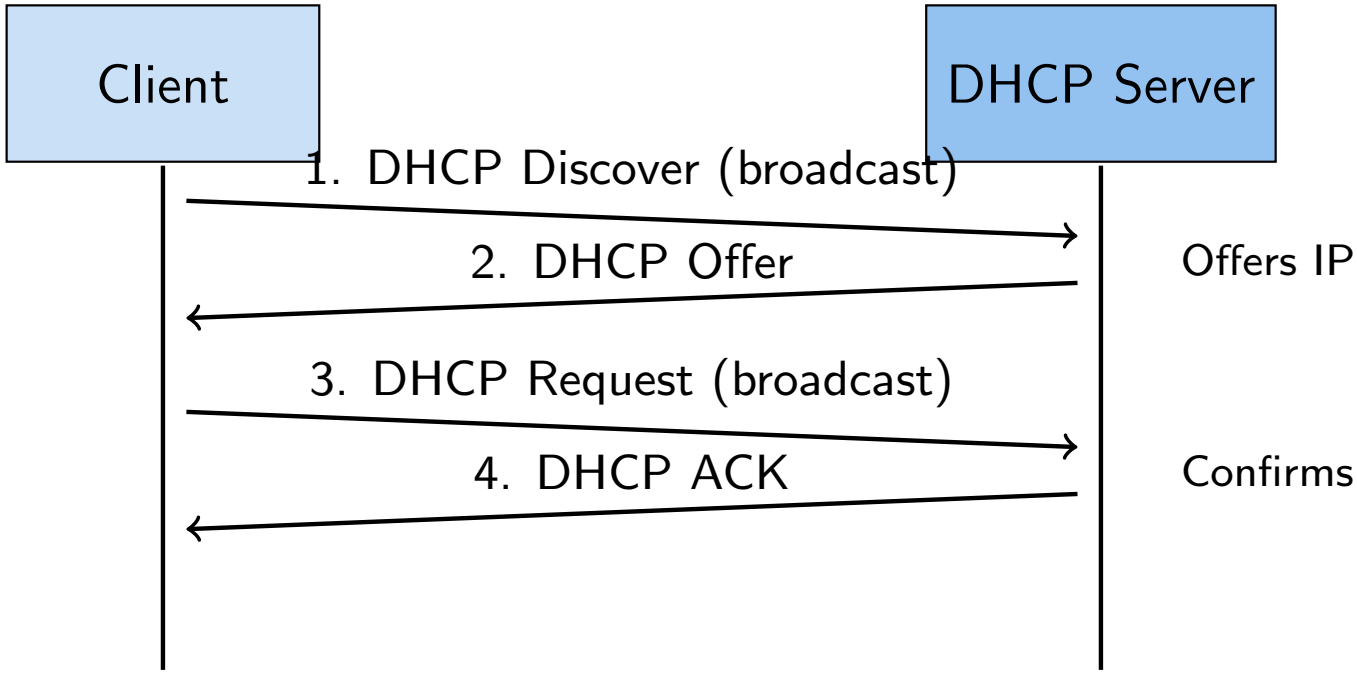
DHCP automatically assigns IP addresses and network configuration to hosts when they join a network. (application layer)

DHCP Provides:

- IP address for the host
- Subnet mask
- Default gateway (router) address
- DNS server address
- Lease time

Why needed? Manual configuration doesn't scale. Imagine configuring thousands of devices manually!

DHCP – Four-Step Process (DORA)



1. **Discover:** Client broadcasts "I need an IP address"
2. **Offer:** Server offers an available IP address
3. **Request:** Client requests the offered address
4. **ACK:** Server confirms the assignment

ICMP – Internet Control Message Protocol

What is ICMP?

ICMP is used by network devices to send error messages and operational information. (network layer, over/part of IP)

Common ICMP Message Types:

Type	Name	Description
0	Echo Reply	Response to ping
3	Destination Unreachable	Can't reach destination
8	Echo Request	Ping request
11	Time Exceeded	TTL expired (traceroute)

Common Uses:

- `ping` – uses Echo Request (8) and Echo Reply (0)
- `traceroute` – uses Time Exceeded (11) messages

Network Layer vs Link Layer

Network Layer (L3)

Scope: Communication between any two network hosts

- Might be on different networks
- Uses **IP addresses**
- Handles **routing**
- Protocols: IP, ICMP, DHCP

Link Layer (L2)

Scope: How packets are sent across individual links

- Same network/LAN
- Uses **MAC addresses**
- Handles **framing**
- Protocols: Ethernet, ARP

Network Layer vs Link Layer

Two Types of Links:

- **Broadcast** (e.g., Ethernet) – shared medium, all nodes hear all transmissions (note that switches are usually transparent)
- **Point-to-Point** (e.g., PPP) – dedicated link between two nodes

Ethernet

Ethernet Overview

The dominant wired LAN technology – simple, cheap, and evolved to keep pace with speed demands.

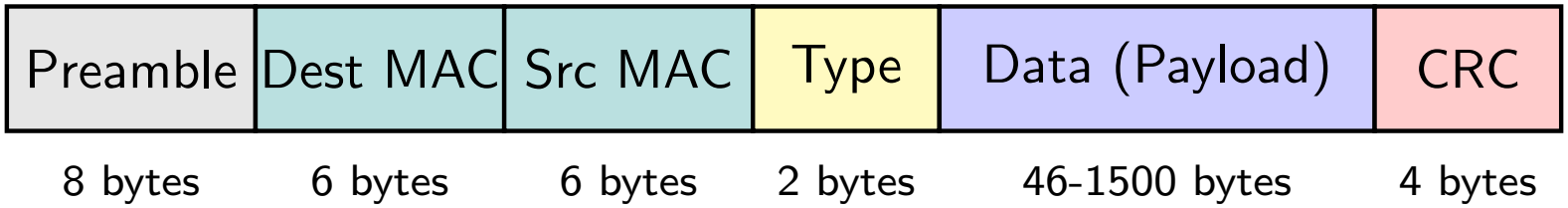
Key Characteristics:

- Interfaces identified using **MAC address** (48 bits)
- Frame begins with **preamble** (0xAAAAAAAB) for synchronization
- Ends with **interpacket gap**
- **CRC** for error detection

Physical Media:

- **Coaxial:** 10Base-5, 10Base-2
- **Twisted Pair (UTP):** 10Base-T, 100Base-TX, 1000Base-T
- **Fiber:** 10GBase-SR, 10GBase-LR

Ethernet Frame Structure



- **Preamble:** 7 bytes of 10101010 + 1 byte of 10101011 (synchronization)
- **Type:** Identifies upper layer protocol (0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6)
- **CRC:** Cyclic Redundancy Check for error detection

MAC Addresses

MAC Address Format

48-bit address burned into NIC ROM (also called physical/LAN/Ethernet address)

Format: `AA:BB:CC:DD:EE:FF` or `AA-BB-CC-DD-EE-FF`

Special Addresses:

- `FF:FF:FF:FF:FF:FF` – Broadcast (sent to all hosts on LAN)
- First 3 bytes = OUI (Organizationally Unique Identifier) – identifies manufacturer

Question: How do we translate an IP address to a MAC address?
Answer: **ARP** (Address Resolution Protocol)

ARP – Address Resolution Protocol

What is ARP?

ARP translates IP addresses to MAC addresses within a local network.

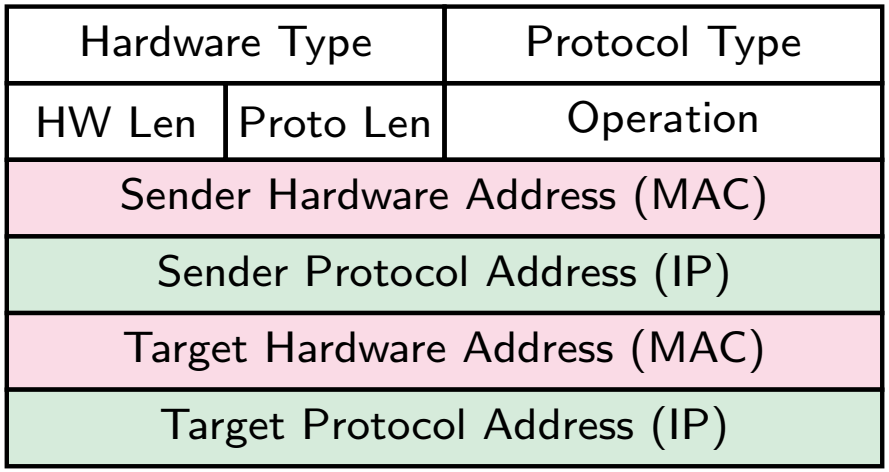
How ARP Works:

1. Host A wants to send to IP 192.168.1.5
2. A broadcasts: **ARP Request** – "Who has 192.168.1.5? Tell 192.168.1.1"
3. Host with that IP responds: **ARP Reply** – "192.168.1.5 is at AA:BB:CC:DD:EE:FF"
4. A caches the mapping in its **ARP table**

View ARP Table:

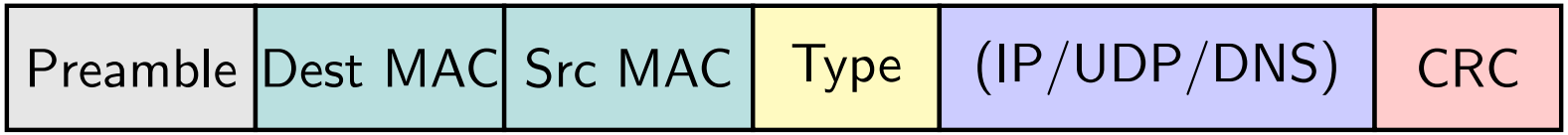
- Linux/Mac: `arp -an`
- Windows: `arp /a`

ARP Packet Structure



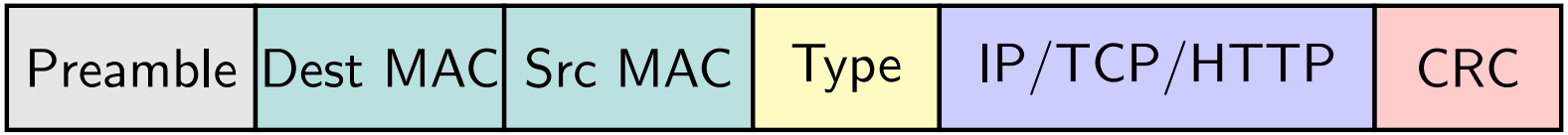
- **Hardware Type:** 1 for Ethernet
- **Protocol Type:** 0x0800 for IPv4
- **Operation:** 1 = ARP Request, 2 = ARP Reply

Protocol Stack – DNS Query



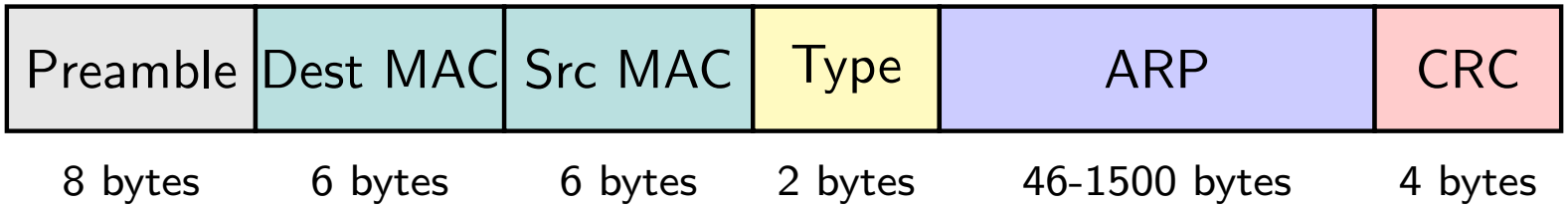
Layer	**Protocol**	**Key Fields**
Application	DNS	Query name, type, class
Transport	UDP	Src port, Dest port (53)
Network	IP	Src IP, Dest IP, TTL
Link	Ethernet	Src MAC, Dest MAC, Type

Protocol Stack – HTTP Request



Layer	**Protocol**	**Key Fields**
Application	HTTP	Method, URL, Headers, Body
Transport	TCP	Src port, Dest port (80), Seq#, Ack#
Network	IP	Src IP, Dest IP, TTL, Protocol=6
Link	Ethernet	Src MAC, Dest MAC, Type=0x0800

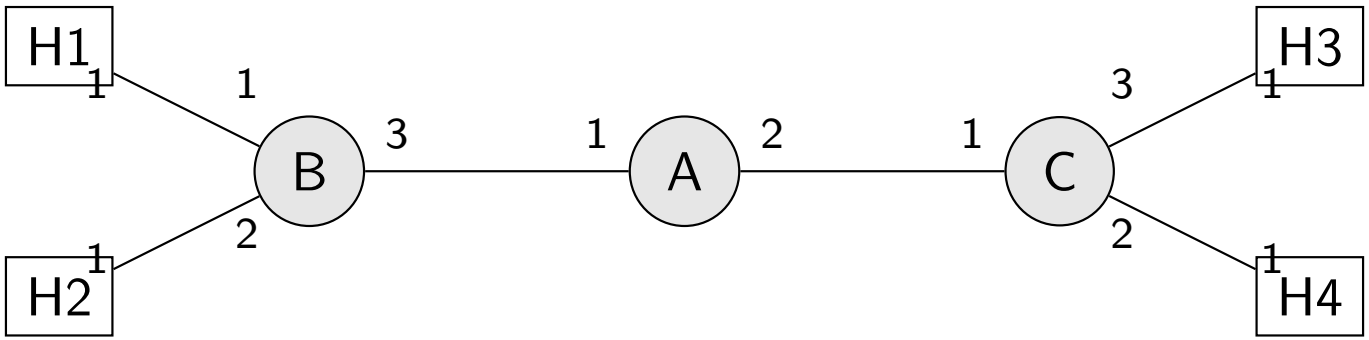
Protocol Stack – ARP



Key Points:

- ARP is **not** an IP protocol – it sits directly on Ethernet
- Ethernet Type field = **0x0806** for ARP
- ARP Request: Dest MAC = `FF:FF:FF:FF:FF:FF` (broadcast)
- ARP Reply: Dest MAC = requester's MAC (unicast)

Exercise 1



A, B, C are routers. Consider sending one IP datagram from **H1 to H4**.

Question: List all link layer frames exchanged until the datagram arrives.
(Assume all ARP tables are initially empty)

Exercise 1 – Solution (1/2)

#	Type	MAC Src	MAC Dest	IP Src	IP Dest
1	ARP	H1	FF-FF-...	–	–
2	ARP	B	H1	–	–
3	IP	H1	B	H1	H4
4	ARP	B	FF-FF-...	–	–
5	ARP	A	B	–	–
6	IP	B	A	H1	H4

Explanation:

1. H1 needs B's MAC → ARP Request (broadcast)
2. B responds with its MAC → ARP Reply
3. H1 sends IP packet to B (first hop)
4. B needs A's MAC → ARP Request
5. A responds → ARP Reply
6. B forwards IP packet to A

Exercise 1 – Solution (2/2)

#	Type	MAC Src	MAC Dest	IP Src	IP Dest
7	ARP	A	FF-FF-...	–	–
8	ARP	C	A	–	–
9	IP	A	C	H1	H4
10	ARP	C	FF-FF-...	–	–
11	ARP	H4	C	–	–
12	IP	C	H4	H1	H4

Key Insight:

- **IP addresses never change** (H1 $\rightarrow$ H4 throughout)
- **MAC addresses change at each hop** (router rewrites)
- Each router must ARP for next hop's MAC

Exercise 2

Setup:

- Host A: IP = 1.1.1.10, MAC = AA-AA-AA-AA-AA-AA
- Host B: IP = 1.1.1.11, MAC = BB-BB-BB-BB-BB-BB
- Both on same local network
- A knows B's IP but not B's MAC

Question: Host A sends **two IP packets** to B immediately after B connects. List all frames exchanged.

Exercise 2

Setup:

- Host A: IP = 1.1.1.10, MAC = AA-AA-AA-AA-AA-AA
- Host B: IP = 1.1.1.11, MAC = BB-BB-BB-BB-BB-BB
- Both on same local network
- A knows B's IP but not B's MAC

Question: Host A sends **two IP packets** to B immediately after B connects. List all frames exchanged.

Solution:

#	Type	MAC Src	MAC Dest	IP Src	IP Dest
1	ARP	AA-AA-...	FF-FF-...	–	–
2	ARP	BB-BB-...	AA-AA-...	–	–
3	IP	AA-AA-...	BB-BB-...	1.1.1.10	1.1.1.11
4	IP	AA-AA-...	BB-BB-...	1.1.1.10	1.1.1.11

Collisions in Ethernet

The Collision Problem

In shared medium (bus topology), multiple nodes may transmit simultaneously $\rightarrow$ **collision**!

CSMA/CD – Carrier Sense Multiple Access with Collision Detection

1. **Carrier Sense:** Listen before transmitting
2. **Collision Detection:** Detect collision while transmitting
3. **Binary Exponential Backoff:** Wait random time before retry

Exponential Backoff Algorithm

After n collisions, choose random K from $\{0, 1, 2, ..., 2^n - 1\}$
Wait $K \times 512$ bit times before retrying
(Maximum: $n = 10$, then $K \in \{0, ..., 1023\}$)

Collision Domain

Collision Domain

A **collision domain** is the network segment where collisions can occur – all nodes that share the same medium.

Reducing Collision Domains:

- **Hubs** – do NOT reduce collision domain (repeat to all ports, transmission is done via single medium)
- **Switches** – each port is its own collision domain! (each port is a dedicated medium)
- **Full-duplex** – eliminates collisions entirely (two connections per port, TX for transmit and RX for receive)

Modern Ethernet: Switches + full-duplex = no collisions! CSMA/CD is effectively obsolete in modern switched networks.

Exercise 3

Question

In Ethernet, after the **7th collision**, what is the probability that a node chooses $K = 7$?

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Question

In Ethernet, after the **7th collision**, what is the probability that a node chooses $K = 7$?

Solution:

After $n = 7$ collisions:

- K is chosen uniformly from $\{0, 1, 2, ..., 2^7 - 1\}$
- That's $\{0, 1, 2, ..., 127\}$ – total of 128 values

$$P(K = 7) = \frac{1}{2^7} = \frac{1}{128}$$

Answer: $\frac{1}{128} \approx 0.78\%$